1 Explanation of a MuJoCo Link Definition

Note: This content was generated by GPT 5.5. Certain parts were augmented with the instructor's notes.

Consider the following MuJoCo XML segment:

```
<body name="link01" pos="0 0 0.0585">
        <inertial pos="2.47e-06 -0.00025198 0.0231717"
          quat="0.708578 0.705633 0.000281462 -0.000355927"
          mass="0.673326"
          diaginertia="0.00128328 0.000839362 0.000719308"/>
        <joint name="joint1" axis="0 0 1"
          range="-2.61799 2.61799"
          actuatorfrcrange="-30 30"/>
        <geom class="visual" mesh="z1_Link01"/>
</body>
```

This script defines a rigid link in a MuJoCo model. In MuJoCo, a robot link is usually represented as a `body`. Each body can contain inertial properties, joints, visual geometries, collision geometries, and other child bodies. The given example defines a body named `link01`, its inertial properties, the joint connecting it to its parent body, and a visual mesh used for rendering.

1.1 Body Definition

The first line is

```
<body name="link01" pos="0 0 0.0585">
```

This creates a MuJoCo body named `link01`. The attribute `name` is simply an identifier that allows this body to be referred to elsewhere in the model.

The attribute

```
pos="0 0 0.0585"
```

defines the position of the body frame with respect to its parent body frame. Therefore, the origin of the `link01` frame is located at

$$^{P}p_B = \begin{bmatrix} 0 \\ 0 \\ 0.0585 \end{bmatrix},$$

where P denotes the parent body frame and B denotes the body frame of `link01`. If the model uses SI units, this means that the origin of `link01` is $0.0585\,\mathrm{m}$ above the parent frame along the parent frame's z-axis.

No orientation is specified for this body. Therefore, the body frame has the same orientation as its parent frame by default.

1.2 Inertial Properties

The inertial properties are defined by

```
<inertial pos="2.47e-06 -0.00025198 0.0231717"
  quat="0.708578 0.705633 0.000281462 -0.000355927"
  mass="0.673326"
  diaginertia="0.00128328 0.000839362 0.000719308"/>
```

This block describes the mass, center of mass, and rotational inertia of the link.

1.2.1 Center of Mass Position

The attribute

```
pos="2.47e-06 -0.00025198 0.0231717"
```

gives the position of the center of mass relative to the `link01` body frame. Thus,

$$ {}^{B}p_{\mathrm{CoM}} = \begin{bmatrix} 2.47 \times 10^{-6} \\ -2.5198 \times 10^{-4} \\ 0.0231717 \end{bmatrix}. $$

This means that the center of mass is almost on the local z-axis of the body frame and is located approximately 2.32 cm above the body-frame origin.

1.2.2 Inertial Frame Orientation

The attribute

```
quat="0.708578 0.705633 0.000281462 -0.000355927"
```

defines the orientation of the inertial frame relative to the body frame. MuJoCo uses the quaternion ordering

$$ q = (w, x, y, z). $$

Therefore, in this example,

$$ q = (0.708578,\ 0.705633,\ 0.000281462,\ -0.000355927). $$

The inertial frame is centered at the center of mass of the body. Its axes are usually aligned with the principal axes of inertia. This is why the inertia matrix can be given in diagonal form using `diaginertia`.

The inertial frame should not be confused with the body frame. The body frame is used to define the kinematic location of the link in the robot structure. The inertial frame is used to express the mass distribution of the link conveniently.

Note: Here "inertial frame" is not used for describing the global frame. MuJoCo uses this term to express the principle axis of rotational inertia for that particular link.

1.2.3 Mass

The attribute

```
mass="0.673326"
```

defines the mass of the body. Therefore,

$$ m = 0.673326. $$

If SI units are used, this corresponds to

$$ m = 0.673326\,\mathrm{kg}. $$

1.2.4 Principal Moments of Inertia

The attribute

```
diaginertia="0.00128328 0.000839362 0.000719308"
```

defines the diagonal inertia matrix of the link with respect to the inertial frame (**note: it means the principle axis, not the global frame**). Therefore,

$$ {}^{I}I = \begin{bmatrix} 0.00128328 & 0 & 0 \\ 0 & 0.000839362 & 0 \\ 0 & 0 & 0.000719308 \end{bmatrix}. $$

Here, I denotes the inertial frame. If SI units are used, the units of these quantities are $\mathrm{kg\,m^2}$.

Since this matrix is diagonal, the products of inertia are zero in the inertial frame. This does not necessarily mean that the inertia matrix is diagonal in the body frame. If the inertia tensor is needed in the body frame, it can be computed as

$$^{B}I = {}^{B}R_I \, {}^{I}I \, \left({}^{B}R_I\right)^T ,$$

where $^{B}R_I$ is the rotation matrix corresponding to the quaternion given in the `quat` attribute.

1.3 Joint Definition

The joint is defined by

```
<joint name="joint1" axis="0 0 1"
  range="-2.61799 2.61799"
  actuatorfrcrange="-30 30"/>
```

This line defines the joint that connects `link01` to its parent body.

1.3.1 Joint Name

The attribute

```
name="joint1"
```

assigns the name `joint1` to this joint. This name can be used later when defining actuators, sensors, or when accessing the joint state in simulation.

1.3.2 Joint Type

No explicit joint type is given. In MuJoCo, if the joint type is not specified, the default joint type is usually a hinge joint, unless changed through a default class elsewhere in the XML file.

A hinge joint is a one-degree-of-freedom revolute joint. Therefore, this joint allows the body `link01` to rotate relative to its parent body.

1.3.3 Joint Axis

The attribute

```
axis="0 0 1"
```

defines the axis of rotation of the hinge joint. This means that the joint rotates about the local z-axis of the body frame.

Therefore, the angular velocity contribution of this joint is in the direction

$$a = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} .$$

For a revolute joint, the joint angular velocity is

$$\omega_1 = \dot{q}_1 a = \dot{q}_1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} .$$

Since no joint position is explicitly specified, the joint is located at the body-frame origin by default. Therefore, the hinge axis passes through the origin of the `link01` body frame.

1.3.4 Joint Range

The attribute

```
range="-2.61799 2.61799"
```

defines the lower and upper limits of the joint coordinate. Therefore,

$$-2.61799 \leq q_1 \leq 2.61799.$$

If the model uses radians, this corresponds approximately to

$$-150^\circ \leq q_1 \leq 150^\circ.$$

This is because

$$2.61799 \text{ rad} \approx 150^\circ.$$

The actual interpretation of the numbers depends on the global angle convention specified in the MuJoCo XML compiler settings. Internally, MuJoCo uses radians.

1.3.5 Actuator Force Range

The attribute

```
actuatorfrcrange="-30 30"
```

limits the actuator-generated generalized force at this joint. Since this is a hinge joint, the generalized force is a torque. Therefore, this range corresponds to

$$-30 \leq \tau_1 \leq 30.$$

If SI units are used, this means

$$-30\,\mathrm{N\,m} \leq \tau_1 \leq 30\,\mathrm{N\,m}.$$

This is not necessarily the input command range of the motor. Rather, it limits the resulting actuator force or torque applied through the joint.

1.4 Visual Geometry

The visual geometry is defined by

```
<geom class="visual" mesh="z1_Link01"/>
```

This line attaches a geometry to the body `link01`. The geometry uses the mesh named `z1_Link01`. This mesh should be defined elsewhere in the XML file, usually inside the `asset` section.

The attribute

```
class="visual"
```

means that this geometry inherits properties from a MuJoCo defaults class named `visual`. In many robot models, visual geometries are used only for rendering and not for collision detection. For example, the `visual` class may define properties such as

```
contype="0"
conaffinity="0"
```

to disable contact generation. However, the actual meaning depends on the definition of the `visual` class elsewhere in the XML file.

The attribute

```
mesh="z1_Link01"
```

specifies which mesh is used to display this link. This mesh affects the appearance of the robot in simulation. Since the body already has an explicit `inertial` element, the mesh does not determine the mass or inertia of this link in this particular definition.

1.5 Summary

In summary, this XML segment defines a rigid link called `link01`. Its body frame is located relative to the parent frame as

$$
{}^{P}p_B = \begin{bmatrix} 0 \\ 0 \\ 0.0585 \end{bmatrix}.
$$

Its center of mass is located relative to the body frame as

$$
{}^{B}p_{\text{CoM}} = \begin{bmatrix} 2.47 \times 10^{-6} \\ -2.5198 \times 10^{-4} \\ 0.0231717 \end{bmatrix}.
$$

Its mass is

$$
m = 0.673326,
$$

and its inertia matrix is diagonal in the inertial frame (**note: principle axis of rotational inertia**):

$$
{}^{I}I = \operatorname{diag}\left(0.00128328,\ 0.000839362,\ 0.000719308\right).
$$

The link is connected to its parent by a revolute joint named `joint1`. This joint rotates about the local z-axis:

$$
a = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.
$$

The joint range is approximately

$$
-150^\circ \leq q_1 \leq 150^\circ,
$$

and the actuator torque is limited to

$$
-30\,\mathrm{N\,m} \leq \tau_1 \leq 30\,\mathrm{N\,m}.
$$

Finally, the link is rendered using the mesh `z1_Link01`, inherited from the `visual` geometry class.